Course: SWE 4304 – Software Project Lab I

DORMATRIX

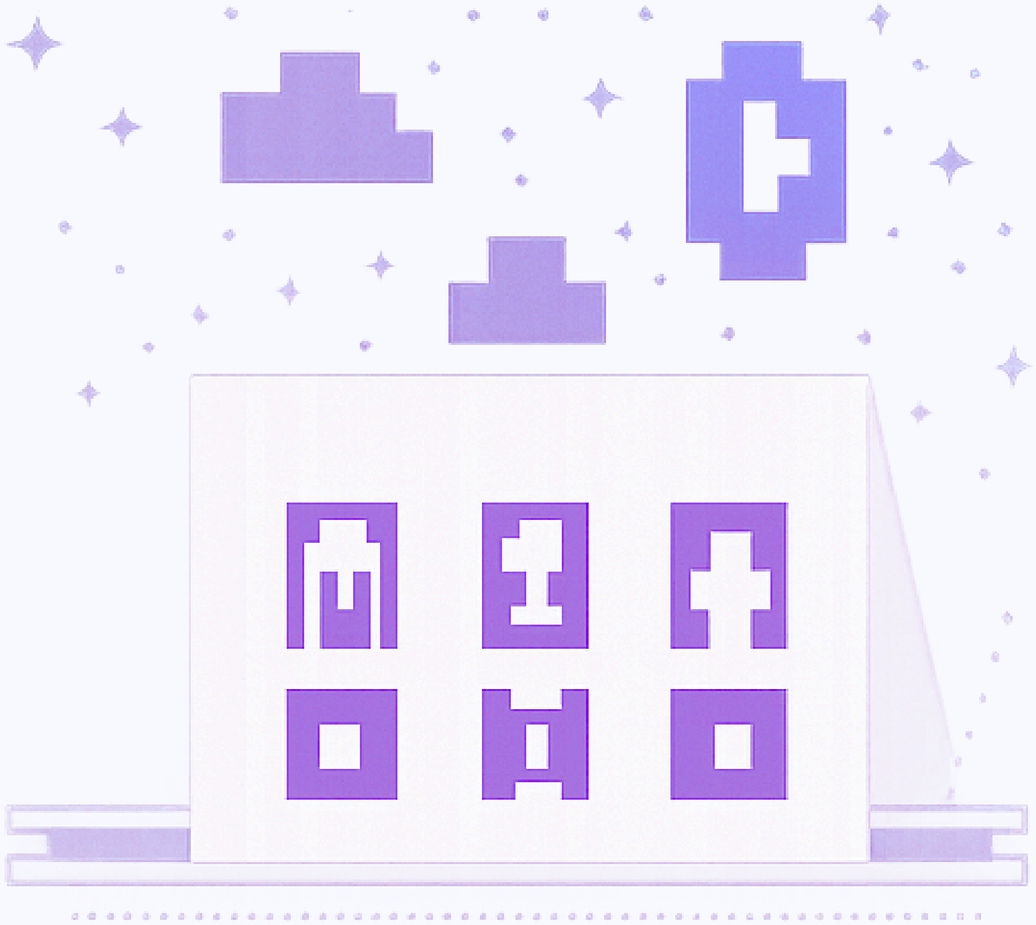
A CLI-Based Dormitory Management System

```
Dormatrix Terminal

C:\Dormatrix> run.bat

            WELCOME TO IUT FEMALE DORMITORY
              Select your role to continue

  [1] Student              [5] Hall Office
  [2] Attendant            [6] Admin
  [3] Maintenance Worker   [7] Cafeteria Manager
  [4] Store In Charge      [0] Exit
```



Final Presentation

Team-1

Team Members

Ayman Binta Altaf Nondiny - 230042141

Khadiza Sultana - 230042135

Procheta Silvie - 230042114

Sayma Tasnim - 230042139

Brief Overview

Project Overview

Dormatrix is a Dormatory Management System consisting of:

1. A role-based CLI dormitory management system
2. 7 user roles each with its own dashboard
3. File-based persistence no database required
4. Built for the IUT Female Dormitory

Problem Statements

- Disoriented complaint resolution
- Inconvenient meal management
- Poor appliance management
- Inadequate hall annoucements
- Unorganized credit tracking
- Lack of lost and found system

Solutions

- Complaint handling
- Food and token management
- Shared facility booking system
- Announcement Dashboard
- Store record keeping
- Centralized lost and found system

Multiple User Views in Dormatrix

1. Students
2. Hall Attendant
3. Maintenance Staff
4. Store In Charge
5. Hall Officer
6. Admin
7. Cafeteria Manager

Features

Implemented Features

1	Authentication and Role Based Access Control
2	Schedule Based Complain Management System
3	Cafeteria Management System
4	Facility Booking System
5	Store Credit And Due System
6	Room Management System

Authentication

1. 7 roles, each with its own user file
2. Passwords hashed via 3-channel DJB2+XOR
3. Masked input with JLine3
4. Controlled access to system features based on assigned role

Dormatrix Terminal

```
C:\Dormatrix> login
Select Role:
> 1
User ID : 230042135
Password: ••••••••
   Login successful
Role: Student
```

Room Management

Student View

1. Room Number, Capacity
2. Application for Room Change
3. View Room Complaints
4. Track Room Change Application

Hall Office View

1. Add Room
2. View Live Occupancy
3. Assign Rooms to Students
4. Solve Room Change Applications

Complaint Management

1. Category Auto Routes to Worker Field Electricity → Electrician
2. Scheduling Checks Student Routine to avoid conflicts
3. Visit Injected on Students' Routine
4. Reshchedule Application by Student when needed

Submitted
↓
Assigned
↓
In Progress
↓
Resolved

Facitilty Booking

Study Room

- Time-slotted Seat Reservation
- Bag-on-Seat Prevention via Warden Timer
- Real-Time Availability Tracking

Laundry

- Automated Wash-Cycle Timer
- Per-Student Double-Booking Prevention
- Timestamp-Based Expiry on Restart

Fridge

- First-free Memory Allocation Algorithm
- Custom Exception Handling
- Ownership Identification System

Cafeteria Meal Tokens

Meal Windows

Meal	Normal	Weekend
Breakfast	7:00-9:30 am	7:00-10:00 am
Lunch	12:00-2:00 pm	12:00-2:00 pm
Dinner	7:00-10:00 pm	7:00-10:00 pm

1. One active token per meal per student
2. Cafeteria Manager Verifies Meal Token
3. Token statuses: Active → Used / Expired

Ramadan: Suhoor, Iftar, Dinner

Store And Balance

Departmental store

1. Inventory listing
2. Recording student purchases
3. Basic dues ledger

Browse Inventory
↓
Add To Cart
↓
Checkout
↓
Sale Record Written
↓
Due Logged if Insufficient

Other Features

1	Lost and Found Module
2	Announcement Module
3	Emergency Contacts
4	Routines
5	Profile Editing

Tools and Technology

Tools and Technology

- Language : Java
- File Based Storage : text based file (.txt)
- Version Control: Github
- Development Environment: IntelliJ IDEA
- Terminal Emulator: WezTerm

Key Algorithms

- Slot-Allocator (Stop-watch cycle)
- Wareden Timer (Auto-Release)
- Complain policy Engine
- DJB2 Algorithm

Libraries Used

Custom-Built Libraries

- MyArrayList
- MyString
- MyOptional
- HashFunction
- SlotAllocator
- TextFile / Logger

External Libraries

- JLine3
- JUnit 5

Project Timeline

Project Timeline

● Completed

Task List	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Dashboard Foundation	■										
Authentication & User System		■	■								
Room Dashboard & Complaint System		■	■								
Facility Booking System				■	■						
Laundry & Permission Mapping				■	■						
Departmental Store Module						■	■				
Food & Meal E-Token System						■	■				
Lost & Found								■			
Event Announcement System								■			
Organize Controllers/Models									■		
Testing & Debugging										■	
Documentation & Final Presentation											■

Contribution

Contribution

Members	Contributions
Ayman Binte Altaf Nondiny	Complaint Module and Scheduling, Announcements and Routine
Khadiza Sultana	Authentication, Cafeteria and Room Management Module, TUI and Testing
Procheta Silvie	Store Management Module
Sayma Tasnim	Facility Booking Module and Lost & Found Module

Thank You

Github Link

github.com/tayma-06/Dormatrix